GNSS Data Collection and Processing Laboratory

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1 Lab Introduction

Global Navigation Satellite Systems (GNSS) are a cornerstone of modern positioning, navigation, and timing (PNT) technologies. They provide accurate location and time information to users anywhere on or near the Earth, under all weather conditions. GNSS has revolutionized many aspects of modern life, providing critical data for navigation, timing, and a vast array of location-based services.

This lab focuses on the practical application of GNSS, from data collection with a receiver to data processing using both professional software and custom Python scripts. You will learn to collect raw GNSS data and process it to determine a precise position, a technique known as single-point positioning.

Upon successful completion of this lab, you will be able to:

- **Objective 1:** Operate GNSS data collection with u-centera highly intuitive GNSS evaluation softwareand u-blox receivers.
- **Objective 2:** Process raw GNSS data using the open-source RTKLIB software suite to estimate position and velocity.
- **Objective 3:** Develop Python scripts to process GNSS data for position estimation.
- **Objective 4:** Analyze GNSS data, including evaluation of positioning error for a static dataset collected in a challenging urban environment, using Python.



Figure 1: (Left) Module-based GNSS positioning. (Right) Integrated GNSS Positioning.

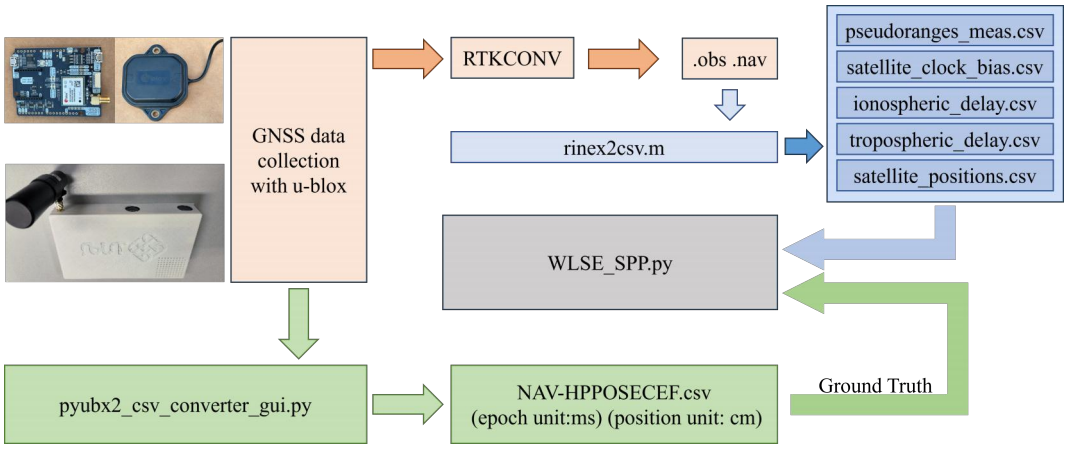


Figure 2: Overview of the lab. Workflow of the data collection and postprocess.

2 Lab Requirements

Lab Requirements

At the end of this laboratory exercise, each group is required to submit a comprehensive lab report. The report should include the following components:

1. **Data Collection:** Description of the data collection process, including hardware setup, software tools used, and environmental conditions.
2. **Raw Data Analysis:** Examination of the raw data collected (e.g., received signals, observed satellites, GDOP, and other quality indicators).
3. **SPP Algorithm:** Explanation of the single-point positioning principle based on least squares estimation (SPP), accompanied by Python code implementation. *Optional:* Explore and design a *weight matrix optimization* strategy to improve positioning accuracy.
4. **Results and Discussion:** Presentation of positioning results, error analysis, and interpretation of findings, including comparison with expectations in urban environments.
5. **Submission Package:** All raw data, the lab report (PDF), Python scripts with documentation, and a README.md file should be organized into a GitHub repository. The repository link must be submitted to both the lecturer and the teaching assistants (TA).

Submission Deadline: TBD

3 Q&A

For questions, technical issues, or further guidance regarding this laboratory exercise, please reach out to the course staff:

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4 GNSS Single Point Positioning (SPP) Theory

This section provides the theoretical foundation for GNSS positioning, including detailed mathematical formulations and practical calculation examples.

4.1 Estimation of Receiver Position and Clock Bias

4.1.1 Basic Pseudorange Observation Model

The fundamental GNSS observation equation for pseudorange measurements is:

$$\rho_i = \sqrt{(x_i^s - x_u)^2 + (y_i^s - y_u)^2 + (z_i^s - z_u)^2} + c \cdot dt_u + I_i + T_i + \epsilon_i \tag{1}$$

where:

- ρ_i = pseudorange measurement to satellite i
- (x_i^s, y_i^s, z_i^s) = satellite i position in ECEF coordinates
- (x_u, y_u, z_u) = user position in ECEF coordinates
- c = speed of light $(299,792,458$ m/s)
- dt_u = receiver clock bias
- I_i = ionospheric delay
- T_i = tropospheric delay
- ϵ_i = measurement noise and multipath error

4.1.2 Linearization for Least Squares Solution

For the iterative least squares solution, we linearize the observation equation around an approximate position (x_0, y_0, z_0):

$$\Delta\rho_i = \frac{\partial \rho_i}{\partial x_u}\Delta x + \frac{\partial \rho_i}{\partial y_u}\Delta y + \frac{\partial \rho_i}{\partial z_u}\Delta z + c \cdot \Delta dt_u \tag{2}$$

The partial derivatives (direction cosines) are:

$$\begin{aligned} a_{xi} &= \frac{\partial \rho_i}{\partial x_u} = -\frac{x_i^s - x_0}{r_i^0} \\ a_{yi} &= \frac{\partial \rho_i}{\partial y_u} = -\frac{y_i^s - y_0}{r_i^0} \\ a_{zi} &= \frac{\partial \rho_i}{\partial z_u} = -\frac{z_i^s - z_0}{r_i^0} \end{aligned} \tag{3}$$

where $r_i^0 = \sqrt{(x_i^s - x_0)^2 + (y_i^s - y_0)^2 + (z_i^s - z_0)^2}$

4.2 Weighted Least Squares Estimation

4.2.1 Design Matrix and Weight Matrix

For n visible satellites, the partial derivatives matrix $\mathbf{H}$ and observation vector $\mathbf{y}$ are:

$$\mathbf{H} = \begin{bmatrix} a_{x1} & a_{y1} & a_{z1} & 1 \\ a_{x2} & a_{y2} & a_{z2} & 1 \\ \vdots & \vdots & \vdots & \vdots \\ a_{xn} & a_{yn} & a_{zn} & 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \Delta\rho_1 \\ \Delta\rho_2 \\ \vdots \\ \Delta\rho_n \end{bmatrix} \tag{4}$$

The weight matrix $\mathbf{W}$ is typically based on satellite elevation angles:

$$\mathbf{W} = \begin{bmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_n \end{bmatrix}, \quad w_i = \sin^2(\theta_i) \tag{5}$$

where θ_i is the elevation angle of satellite i.

4.2.2 Solution and Covariance

The weighted least squares solution is:

$$\hat{\mathbf{x}} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\mathbf{y} \tag{6}$$

The covariance matrix of the estimated parameters:

$$\mathbf{C}_{\hat{x}} = \sigma_0^2(\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1} \tag{7}$$

where σ_0^2 is the variance of unit weight.

4.3 Practical Calculation Example

4.3.1 Given Data

Consider a scenario with 4 GPS satellites at epoch 2025-09-23 12:00:00 UTC:

Table 1: Example satellite positions and pseudoranges

SV	X (m)	Y (m)	Z (m)	ρ (m)
G01	15,642,391.2	-18,234,567.8	9,876,543.2	23,456,789.1
G05	-12,345,678.9	20,123,456.7	11,234,567.8	22,876,543.2
G12	8,765,432.1	-15,678,901.2	19,012,345.6	24,123,456.7
G15	-19,876,543.2	-12,345,678.9	8,901,234.5	23,678,901.2

Initial position estimate: $(x_0, y_0, z_0) = (-2,418,093.0, 5,386,767.0, 2,405,602.0)$ (Hong Kong)

4.3.2 Step 1: Calculate Geometric Ranges

For each satellite, compute the geometric range:

$$r_i^0 = \sqrt{(x_i^s - x_0)^2 + (y_i^s - y_0)^2 + (z_i^s - z_0)^2} \tag{8}$$

Example for G01:

$$r_1^0 = \sqrt{(15,642,391.2 - (-2,418,093.0))^2 + (-18,234,567.8 - 5,386,767.0)^2} \tag{9}$$

$$+ \sqrt{(9,876,543.2 - 2,405,602.0)^2} \tag{10}$$

$$= 30,234,567.8 \text{ m} \tag{11}$$

4.3.3 Step 2: Form Design Matrix

Calculate direction cosines for each satellite:

$$\mathbf{H} = \begin{bmatrix} 0.5987 & -0.7812 & 0.2471 & 1 \\ -0.3124 & 0.4897 & 0.3012 & 1 \\ 0.3654 & -0.7123 & 0.5498 & 1 \\ -0.5234 & -0.2876 & 0.1987 & 1 \end{bmatrix} \tag{12}$$

4.3.4 Step 3: Calculate Observation Vector

$$\mathbf{y} = \begin{bmatrix} 23,456,789.1 - 30,234,567.8 - I_1 - T_1 \\ 22,876,543.2 - 29,123,456.7 - I_2 - T_2 \\ 24,123,456.7 - 28,987,654.3 - I_3 - T_3 \\ 23,678,901.2 - 29,876,543.2 - I_4 - T_4 \end{bmatrix} \tag{13}$$

With typical corrections: $I_i \approx 5$ m, $T_i \approx 2.3$ m

4.3.5 Step 4: Solve for Position and Clock Bias Updates

Using weighted least squares:

$$\hat{\mathbf{x}} = \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \\ c \cdot \Delta dt_u \end{bmatrix} = \begin{bmatrix} 12.4 \\ -8.7 \\ 5.2 \\ -6,234.5 \end{bmatrix} \tag{14}$$

4.3.6 Step 5: Update Position and Clock Bias

$$x_{new} = x_0 + \Delta x = -2,418,093.0 + 12.4 = -2,418,080.6 \text{ m} \tag{15}$$

$$y_{new} = y_0 + \Delta y = 5,386,767.0 - 8.7 = 5,386,758.3 \text{ m} \tag{16}$$

$$z_{new} = z_0 + \Delta z = 2,405,602.0 + 5.2 = 2,405,607.2 \text{ m} \tag{17}$$

$$dt_{u,new} = dt_{u,0} + \Delta dt_u / c = 0 - 20.8 \times 10^{-6} = -20.8 \mu\text{s} \tag{18}$$

4.4 Estimation of Receiver Velocity and Clock Drift

4.4.1 Doppler-Based Velocity Estimation

The Doppler shift observation equation:

$$f_D^i = -\frac{f_L}{c} \cdot \left[(\mathbf{v}_i^s - \mathbf{v}_u) \cdot \mathbf{e}_i + c \cdot \dot{dt}_u \right] \tag{19}$$

where:

- f_D^i = Doppler frequency shift for satellite i
- f_L = carrier frequency (L1: 1575.42 MHz)
- $\mathbf{v}_i^s$ = satellite velocity vector
- $\mathbf{v}_u$ = user velocity vector
- $\mathbf{e}_i$ = unit vector from user to satellite
- $\dot{dt}_u$ = receiver clock drift

4.4.2 Velocity Design Matrix

Similar to position estimation, we form the velocity design matrix:

$$\mathbf{H}_v = \begin{bmatrix} e_{x1} & e_{y1} & e_{z1} & 1 \\ e_{x2} & e_{y2} & e_{z2} & 1 \\ \vdots & \vdots & \vdots & \vdots \\ e_{xn} & e_{yn} & e_{zn} & 1 \end{bmatrix} \tag{20}$$

4.4.3 Practical Velocity Calculation Example

Given Doppler measurements at the same epoch:

Table 2: Example Doppler measurements and satellite velocities

SV	f_D (Hz)	v_x^s (m/s)	v_y^s (m/s)	v_z^s (m/s)
G01	-2,345.6	892.3	-2,134.5	1,234.2
G05	1,876.4	-1,567.8	987.6	-567.8
G12	-987.3	2,345.6	1,234.5	-890.1
G15	567.8	-678.9	-1,890.2	2,134.5

4.4.4 Step 1: Convert Doppler to Range Rate

$$\dot{\rho}_i = -\frac{c \cdot f_D^i}{f_L} \tag{21}$$

For G01: $\dot{\rho}_1 = -\frac{299{,}792{,}458 \times (-2{,}345.6)}{1{,}575.42 \times 10^6} = 446.5$ m/s

4.4.5 Step 2: Calculate Relative Velocities

$$\Delta\dot{\rho}_i = \dot{\rho}_i - \mathbf{v}_i^s \cdot \mathbf{e}_i \tag{22}$$

4.4.6 Step 3: Solve for User Velocity and Clock Drift

Using the same weighted least squares approach:

$$\hat{\mathbf{v}} = \begin{bmatrix} v_{x,u} \\ v_{y,u} \\ v_{z,u} \\ c \cdot \dot{dt}_u \end{bmatrix} = \begin{bmatrix} 0.2 \\ -0.1 \\ 0.0 \\ 0.5 \end{bmatrix} \tag{23}$$

This indicates a nearly stationary receiver with small velocity components due to measurement noise.

4.5 Error Budget Analysis

The total positioning error budget for SPP includes:

Table 3: Typical SPP error budget

Error Source	**1-σ Error (m)**	**Mitigation Method**
Satellite clock	1.5	Broadcast corrections
Ephemeris	2.0	Precise ephemeris
Ionosphere	5.0	Dual-frequency or models
Troposphere	0.5	Models (e.g., Saastamoinen)
Multipath	1.0	Signal processing, antenna design
Receiver noise	0.3	High-quality receiver
Total (RSS)	**5.6**	

With typical DOP values (HDOP $\approx$ 1.5, VDOP $\approx$ 2.5), the expected positioning accuracy is:

- Horizontal: $5.6 \times 1.5 = 8.4$ m (95%)
- Vertical: $5.6 \times 2.5 = 14.0$ m (95%)

5 GNSS Data Collection and Processing with u-blox Receiver

This section will guide you through collecting data with a u-blox receiver using the u-center evaluation software and processing it using the RTKLIB software suite.

5.1 U-blox Receiver and u-center Software Overview

The u-center GNSS evaluation software is a powerful, comprehensive tool specifically designed for u-blox positioning products. According to u-blox, u-center offers:

- Full compatibility with u-blox M8, M9, F9, and legacy GNSS products
- Quick product configuration for key use cases
- Flexible user interface with personalized workspaces
- Powerful logging functionality for efficient development support
- Easy evaluation of u-blox GNSS services including AssistNow

The software is **u-center:** Designed for u-blox M8, M9, F9, and legacy GNSS products

We will use a commercial-grade u-blox GNSS receiver kit (e.g., ZED-F9P), which supports multi-constellation and multi-band signals for high-precision positioning.



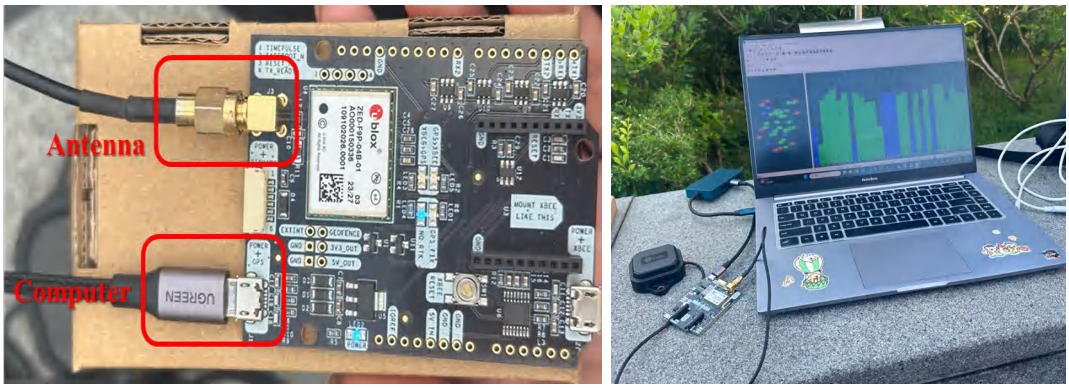
Figure 3: Illustration of the u-blox GNSS receiver kit.

- **Multi-band GNSS Receiver:** Supports concurrent reception of multiple GNSS constellations (GPS, GLONASS, Galileo, BeiDou).
- **Concurrent Multi-Constellation Reception:** Receives signals from various constellations simultaneously, improving accuracy, reliability, and availability, especially in challenging environments.
- **Compact & Power-Efficient:** Ideal for integration into various mobile and static applications.

5.2 Data Collection Procedure

5.2.1 Hardware Setup

Connect the antenna, receiver, and your computer as shown in the figures below. A clear view of the sky is essential for a good signal lock.



(a) u-blox F9P. (b) Computer Connection Settings.

Figure 4: Illustration of the u-blox GNSS receiver F9P kit connection with the antenna.

5.2.2 Using u-center for Data Logging

u-center provides an intuitive interface for GNSS evaluation with personalized workspaces and adaptive window elements. The software's log player provides easy message-based and time-based navigation with adjustable playback speed.

Step 1: Install u-center

Download and install the software from the official u-blox website: https://www.u-blox.com/en/product/u-center. Regular updates are provided to ensure the software has the latest functionalities and pre-defined configurations.

Step 2: Connect the Receiver

Open u-center. From the top menu, select the correct COM port for your connected receiver and set the baud rate (typically 9600 or 115200) to establish communication.

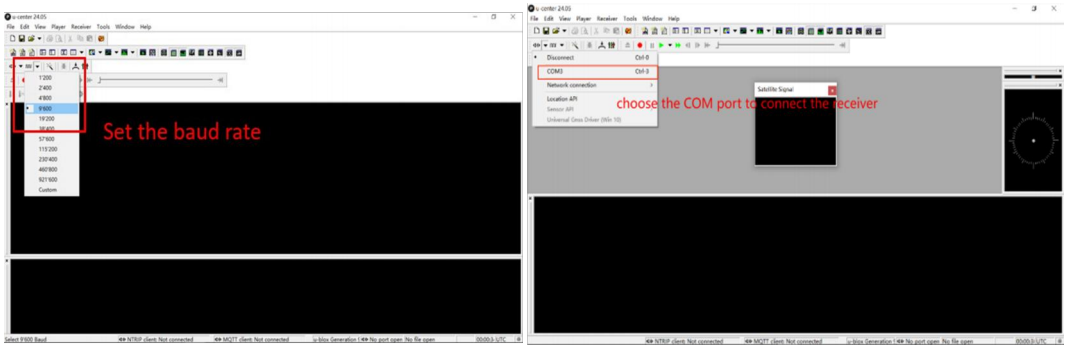


Figure 5: Illustration of the u-center software in the computer: connect receiver

Step 3: Verify Data Stream

Once connected, u-center will display real-time data, including satellite visibility, signal strength (C/N0), and position coordinates. The software supports product evaluation with a choice of views to observe static and dynamic behavior of the connected u-blox GNSS receiver.

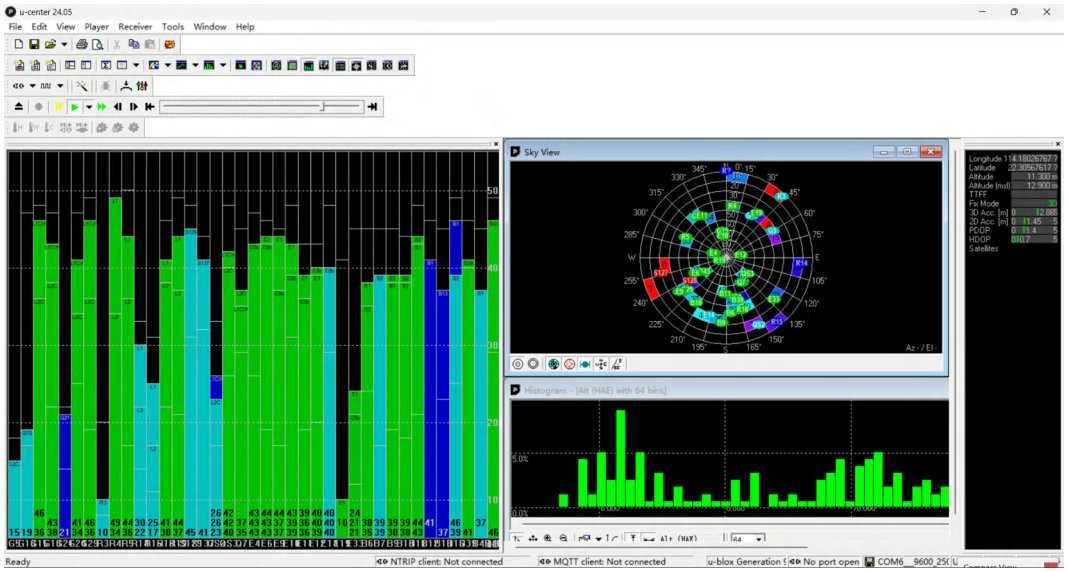


Figure 6: Illustration of the u-center software in the computer with satellite visualization.

Step 4: Start Recording

u-center's powerful logging functionality enables efficient development support. Click the Record button in the menu bar. Choose a location and name for your data file (it will be saved with a .ubx extension).

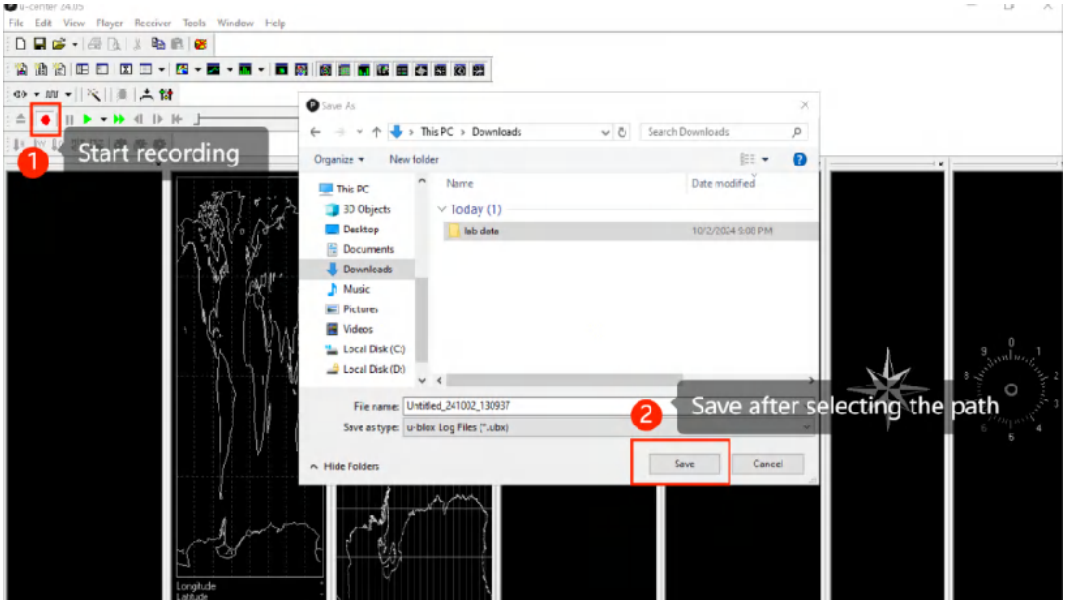


Figure 7: Illustration of the u-center software in the computer: start recording.

Step 5: Stop Recording

Once you have collected sufficient data, click the Stop button in the u-center menu bar. The .ubx file is now saved and ready for processing.

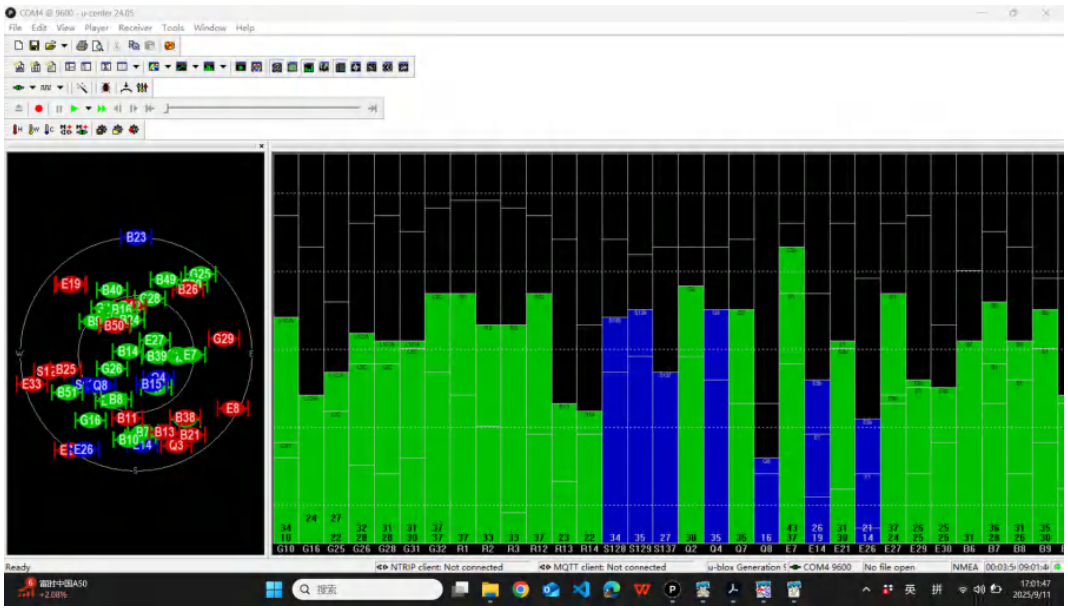


Figure 8: Record data visualization page

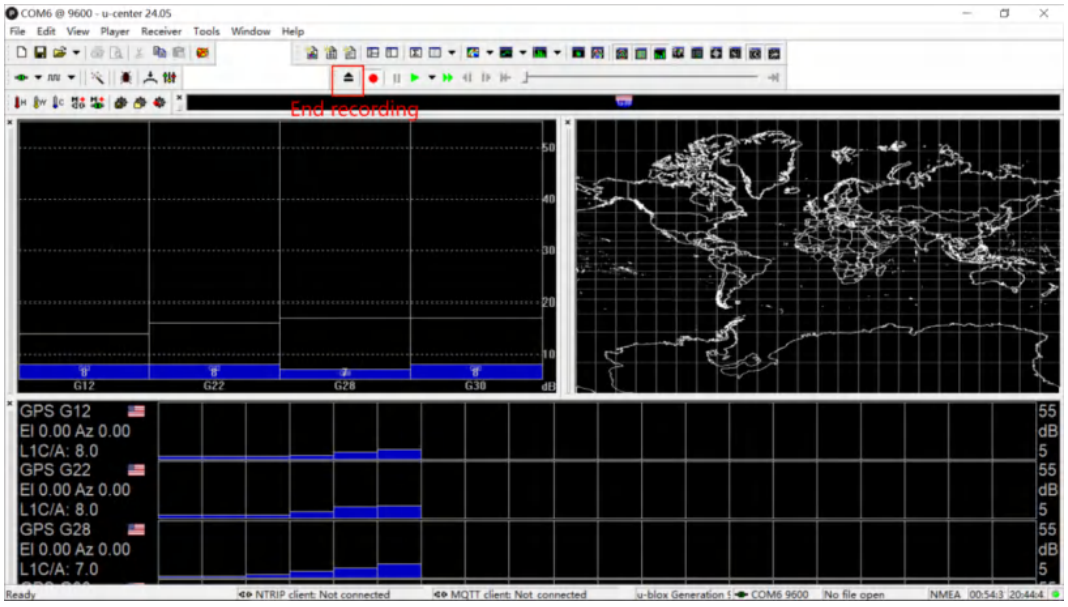


Figure 9: Illustration of the u-center software in the computer: stop recording.

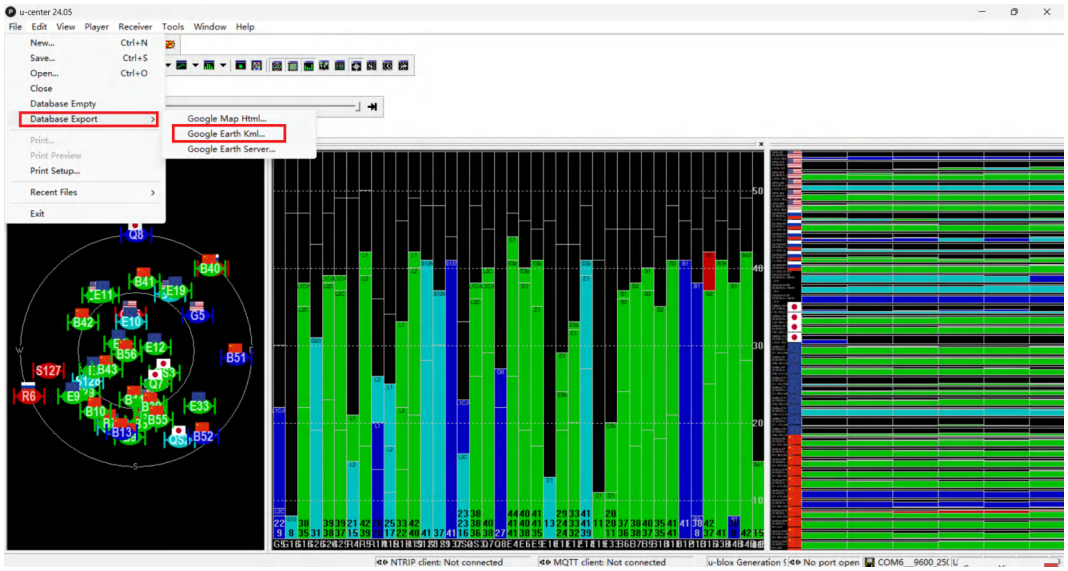


Figure 10: Export the ublox positioning result to kml.

5.2.3 Quick Product Configuration

u-center's quick product configuration feature allows users to define or apply GNSS product configurations for specific use cases. Saving, restoring, or sharing configurations between different products and users is easy, making it ideal for standardized testing scenarios.

5.2.4 Experimental Task: Static Data Collection

The primary goal is to collect a static dataset in a typical urban environment to analyze positioning performance.



Figure 11: Collection location examples

Experimental Site: The Hong Kong Polytechnic University.



Figure 12: Collection environment examples

Experimental Environment: This location is a semi-open area surrounded by tall buildings, creating a challenging "urban canyon" environment where GNSS signals can be blocked or reflected (multipath).

Procedure:

1. Set up the GNSS receiver and antenna at the designated location.
2. Ensure the antenna has the best possible sky view.
3. Begin recording data using u-center as described above.
4. The software enables easy setup and evaluation of u-blox GNSS services. If you are the first time connecting the u-blox kit, please click "Receiver - Debug Meassage" before data collection.

5.3 Data Analysis with RTKLIB

RTKLIB is an open-source software package for standard and precise GNSS positioning.

Step 1: Install RTKLIB
Download the recommended version from the official source: RTKLIB Download.
Step 2: Launch RTKLIB
Unzip the files and run rtklaunch.exe. This opens a control panel for all RTKLIB tools.

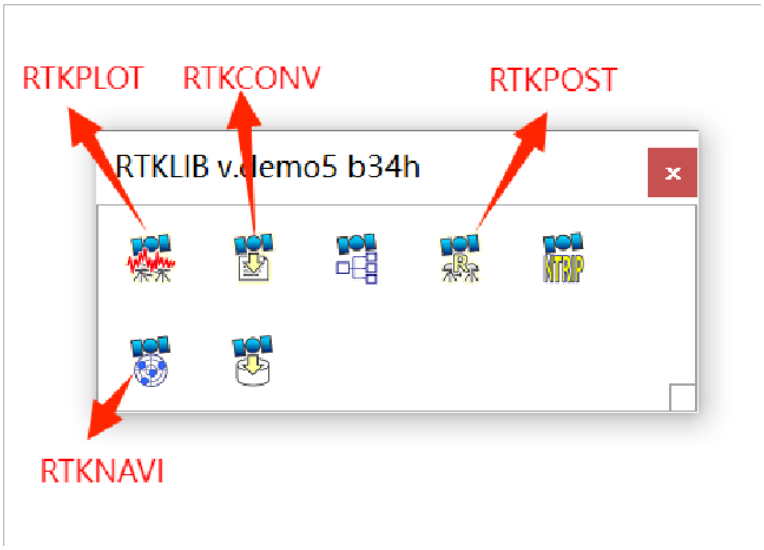


Figure 13: Illustration of the rtklaunch in the computer.

5.3.1 Convert Raw Data to RINEX

RINEX (Receiver Independent Exchange Format) is the standard format for raw GNSS data. We must convert our proprietary .ubx file.

Step 3: Open RTKCONV from the RTKLIB launcher.

- In the first field (labeled "RTCM, RCV RAW or RINEX OBS"), select your .ubx log file.
- Ensure the format is set to u-blox (UBX).
- In the output directories, make sure the .obs (observation) and .nav (navigation/ephemeris) files are checked.
- Click Convert. This will generate the necessary RINEX files.

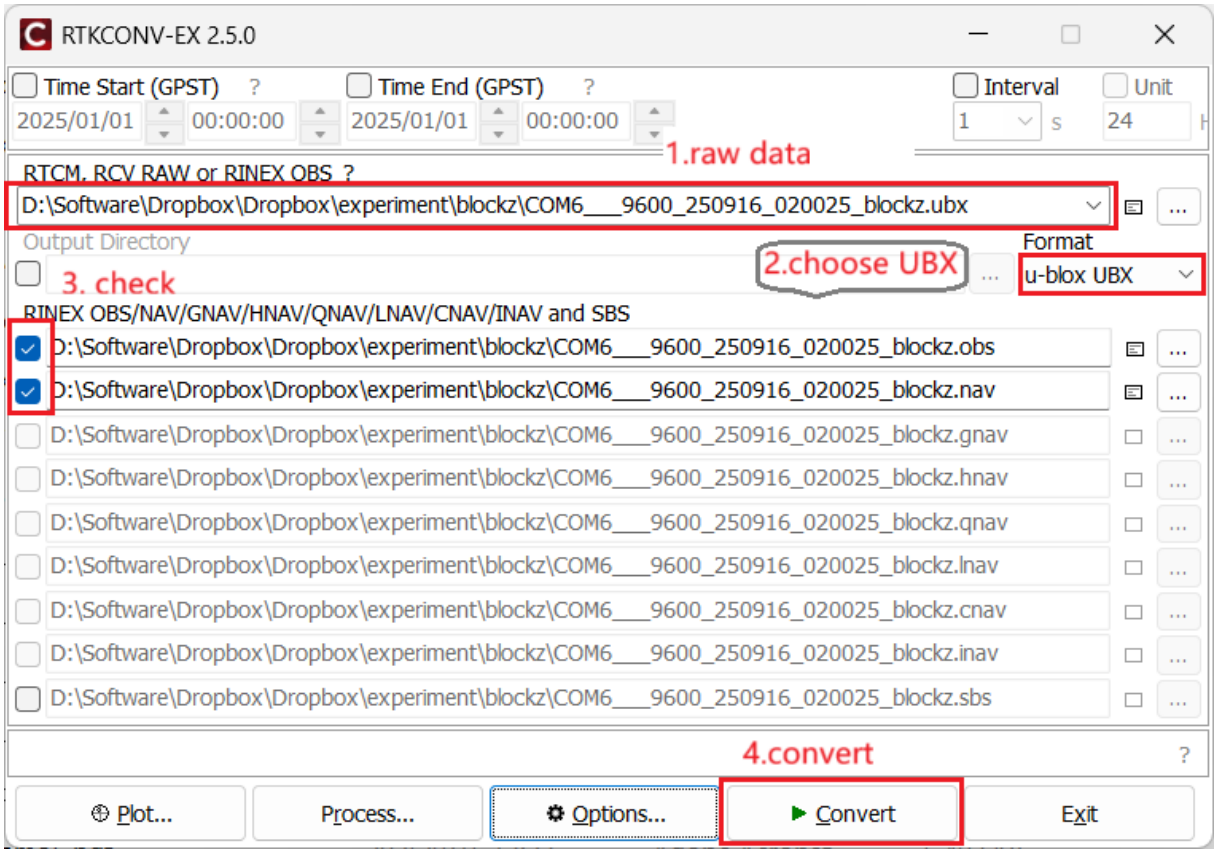


Figure 14: Illustration of the rtkconv software in the computer: convert .ubx to RINEX.

5.3.2 Download Precise Ephemeris Data

We can use SPP to get a coarse trajectory. For better accuracy, we can use precise ephemeris data from a local reference station.

Step 4: Navigate to the Hong Kong Geodetic Survey Services website: HK SatRef RINEX Data.

Step 5: Select the reference station and set the data type as needed.

Step 1. Select the required stations

Hong Kong SatRef Station			
☐**HKKS** Kau Sai Chau	☐**HKKT** Kam Tin	☐**HKLM** Lamma Island	☐**HKLT** Lam Tei
☐**HKMW** Mui Wo	☐**HKNP** Ngong Ping	☐**HKOH** Obelisk Hill	☐**HKPC** Peng Chau
☑**HKSC** Stonecutters Island	☐**HKSL** Siu Lang Shui	☐**HKSS** Shap Sze Heung	☐**HKST** Sha Tin
☐**HKTK** Sha Tau Kok	☐**HKWS** Wong Shek	☐**HKCL** Chek Lap Kok	☐**HKQT** Quarry Bay
☐**T430** Trig 430	☐**HKFN** Fanling	☐**KYC1** Kau Yi Chau	**TCHK*** Tate's Cairn
Macao MoSRef Station			
☐**COAL** Coloane Alto	☐**DSMG** Taipa Grande	☐**FOMO** Monte Fortress	

*Note: 1. Satellite Reference Station summary has been replaced by Satellite Reference Station Coordinates List since 1 Apr, 2016.
2. Data of TCHK is not available

Step 2. Select Data Format

File Length	Interval	Data Format	
		RINEX 2.11	RINEX 3.02
1 Hour	1 Second	**Not Select**	**Selected**
	5 Seconds	**Not Select**	**Not Select**
24 Hours	5 Seconds	**Not Select**	Not Available
	30 Seconds	**Not Select**	**Not Select**

Figure 15: Illustration of step1 and step 2 of downloading ephemeris

Step 6: Open your .obs file (from Step 3) with a text editor and note the start and end times of your data collection.

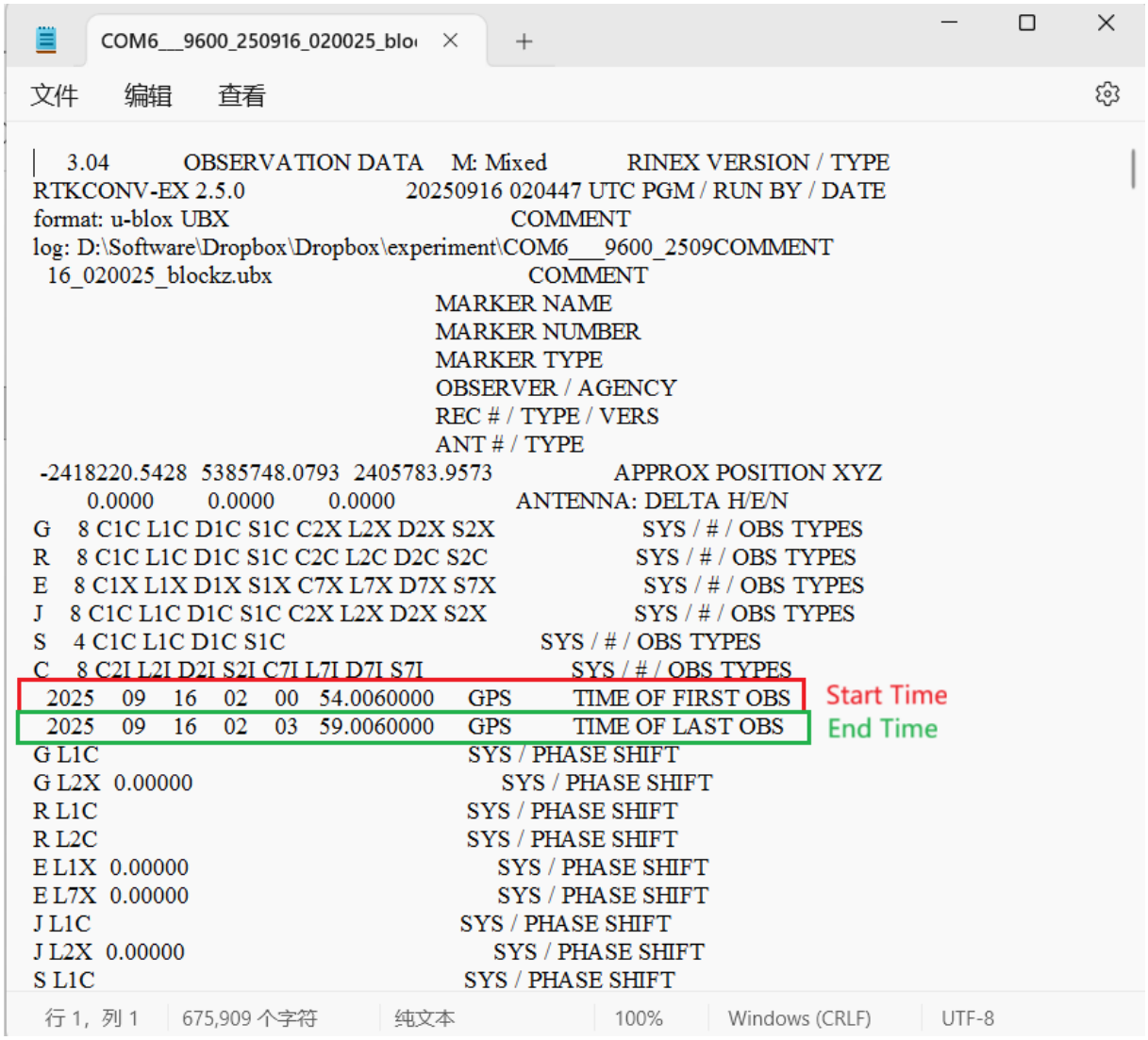


Figure 16: Illustration of the start and End time.

Step 7: On the website, enter the corresponding UTC start and end times for your data collection period and submit.

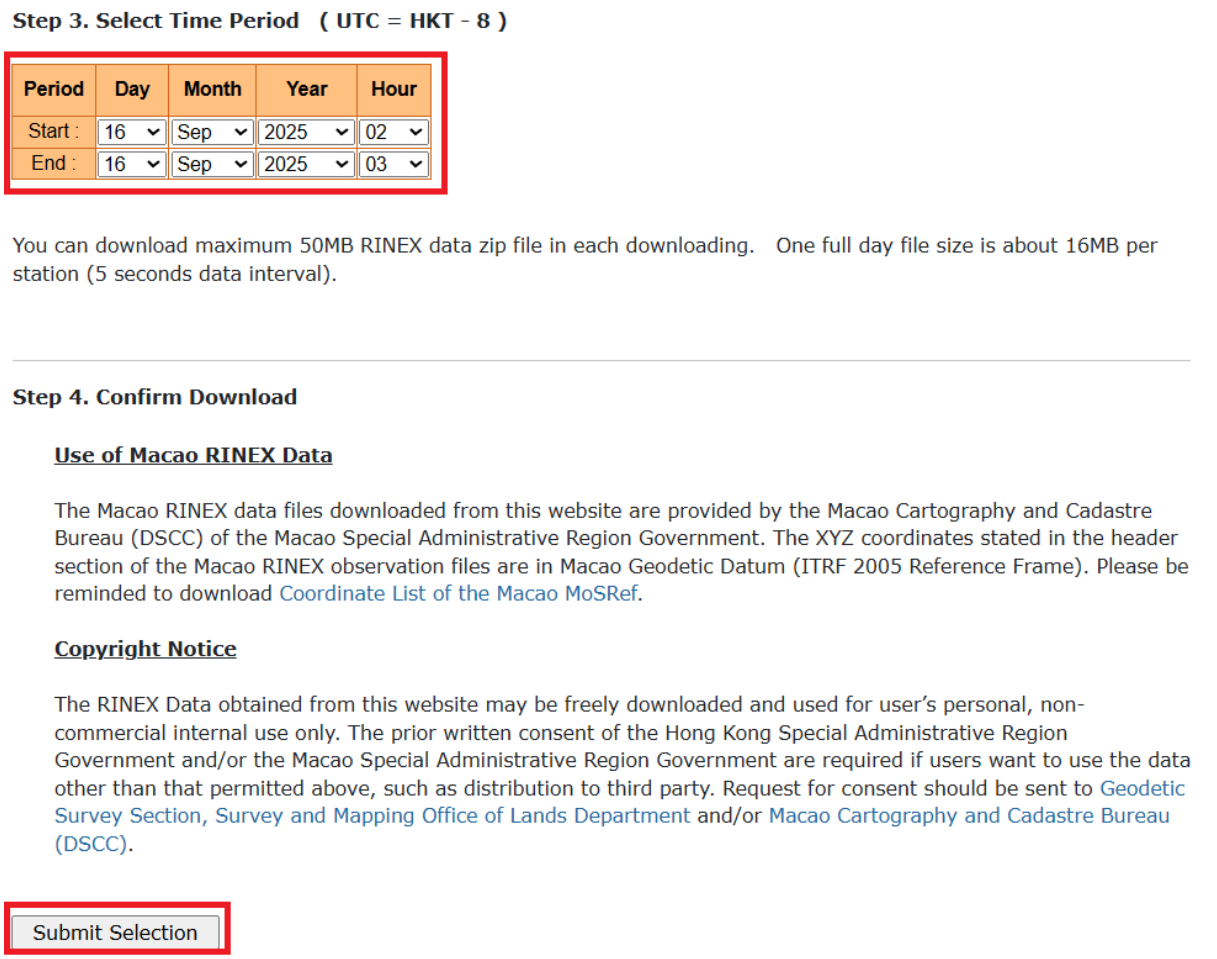


Figure 17: Illustration of step3 and step 4 of downloading ephemeris.



Figure 18: Illustration of how to get output folder.

hksc259c.25f	BEIDOU (CN)	2025/9/28 16:15	25F 文件	114 KB
hksc259c.25g	GLONASS (Russia)	2025/9/28 16:15	25G 文件	14 KB
hksc259c.25l	Galileo (EU)	2025/9/28 16:15	25L 文件	76 KB
hksc259c.25m		2025/9/28 16:15	25M 文件	6 KB
hksc259c.25n	GPS (USA)	2025/9/28 16:15	25N 文件	27 KB
hksc259c.25o	Observations of station	2025/9/28 16:15	25O 文件	28,239 KB
hksc259c.25q	QZSS(Japan)	2025/9/28 16:15	25Q 文件	6 KB

Figure 19: Illustration of ephemeris.

5.3.3 Post-Process for a Position Solution

Now we will use RTKPOST to compute the position.

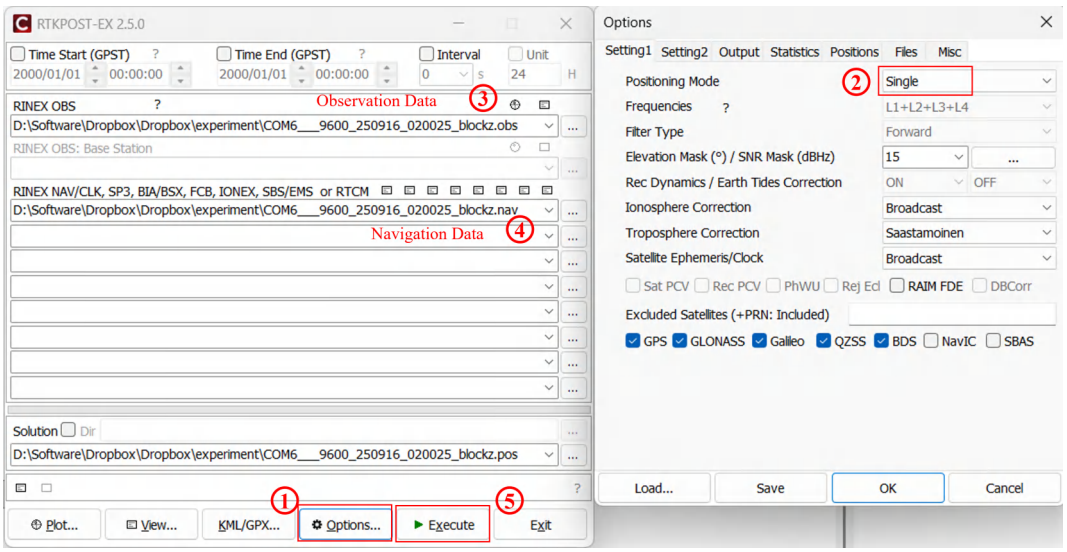


Figure 20: Illustration of the usage of rtkpost for spp solution.

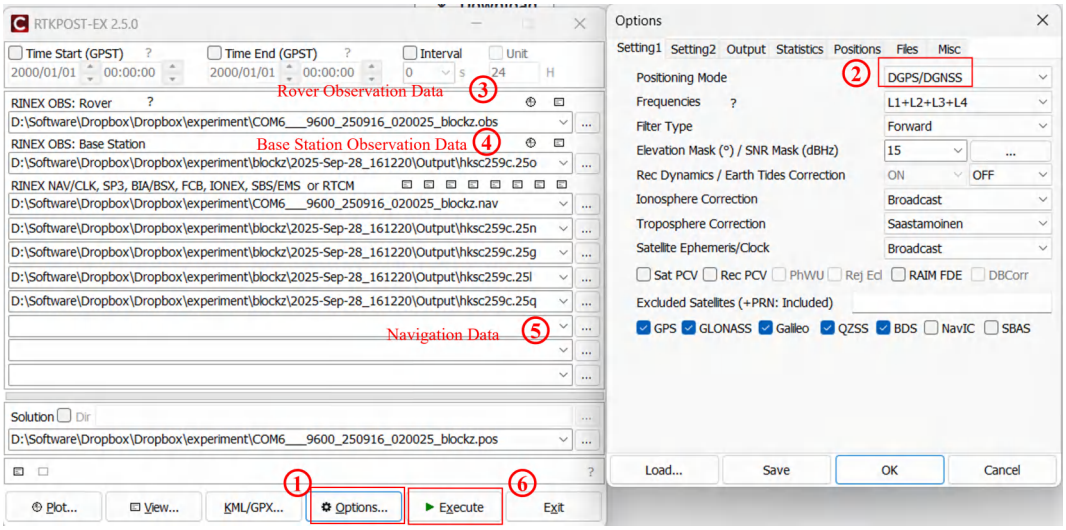


Figure 21: Illustration of the usage of rtkpost for rtk solution.

Step 10: Open RTKPOST from the RTKLIB launcher.

- **RINEX OBS (Rover):** Select your .obs file generated by RTKCONV.
- **RINEX OBS (Base Station):** Select the .25o (or similar) file downloaded from the HK SatRef website.
- **RINEX NAV/CLK:** Select the .nav file generated by RTKCONV.

5.3.4 View and Analyze Results

Step 11: Plot Results

In RTKPOST, click Plot to open RTKPLOT. This tool visualizes the results, showing the ground track, position deviation over time, satellite visibility, and more. For a static test, the ground track should be a tight cluster of points.

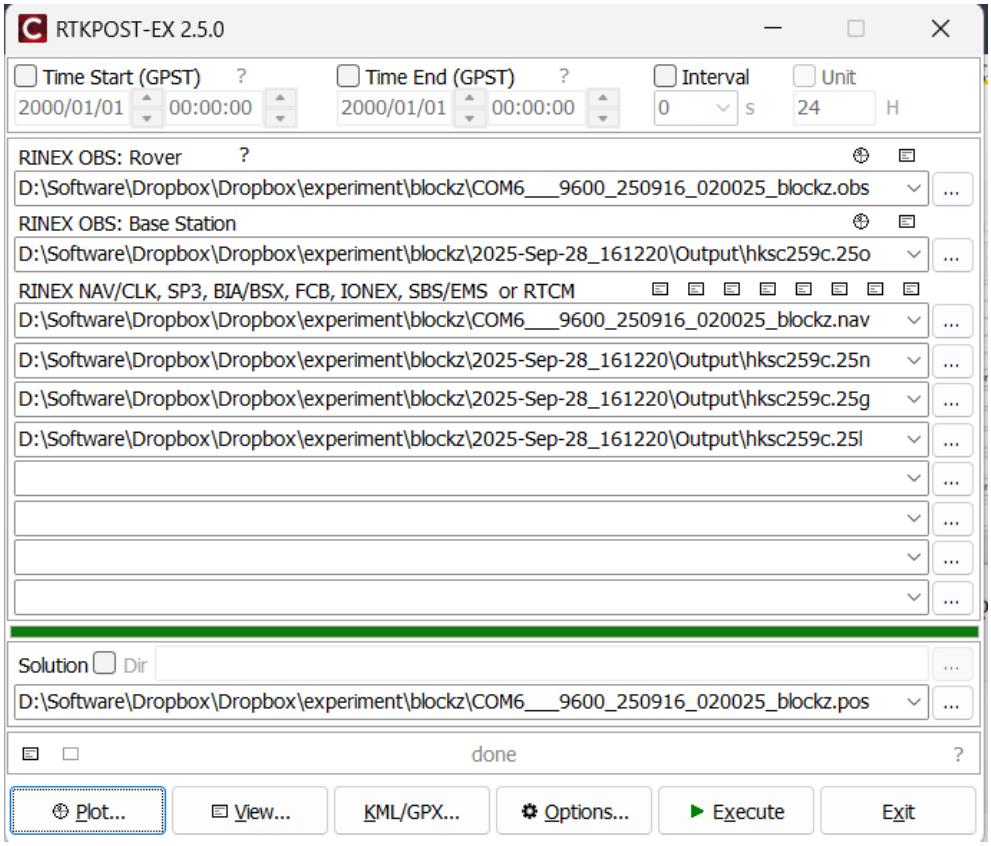


Figure 22: Illustration of plot button in rtkpost.

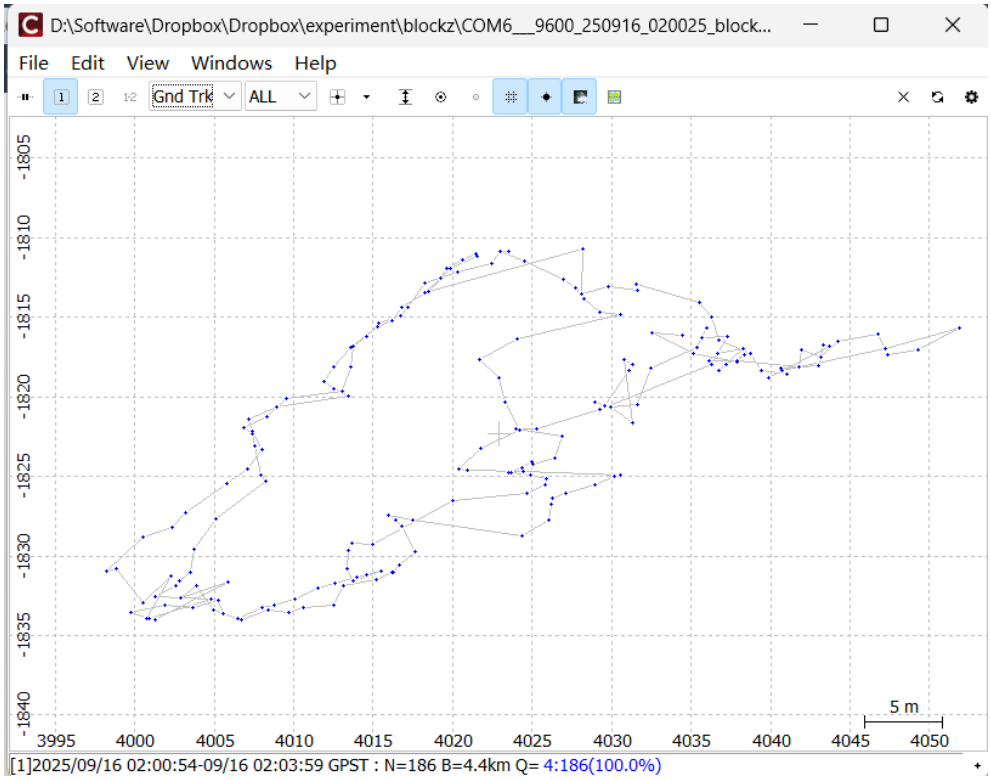


Figure 23: Illustration of the ground track.

5.3.5 Display results in Google Earth

Step 12: View in Google Earth

1. In RTKPOST, go to Tools → Convert to KML. Select your .pos file as input and convert it.
2. Open the Google Earth website.
3. In the menu, go to Projects → New Project → Import KML file from computer and upload your generated KML file to see your calculated track overlaid on the map.

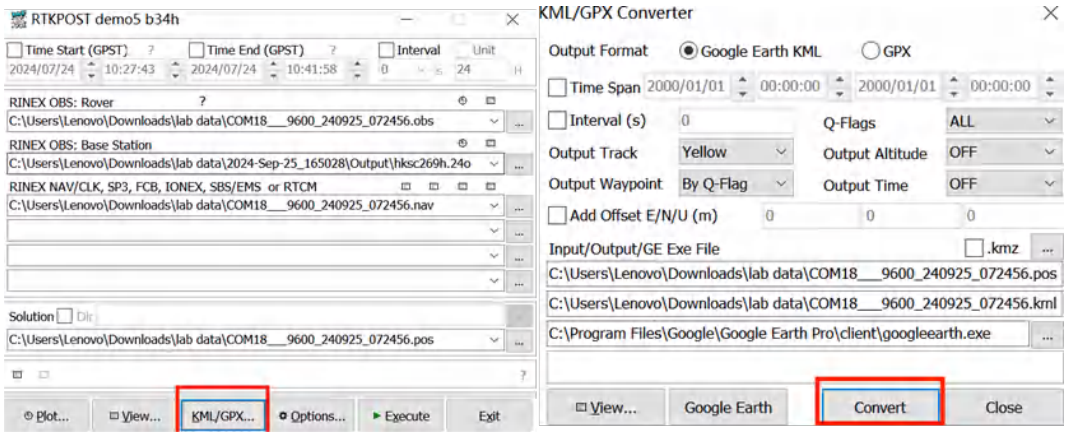


Figure 24: Illustration of rtkpost: convert .pos to KML.

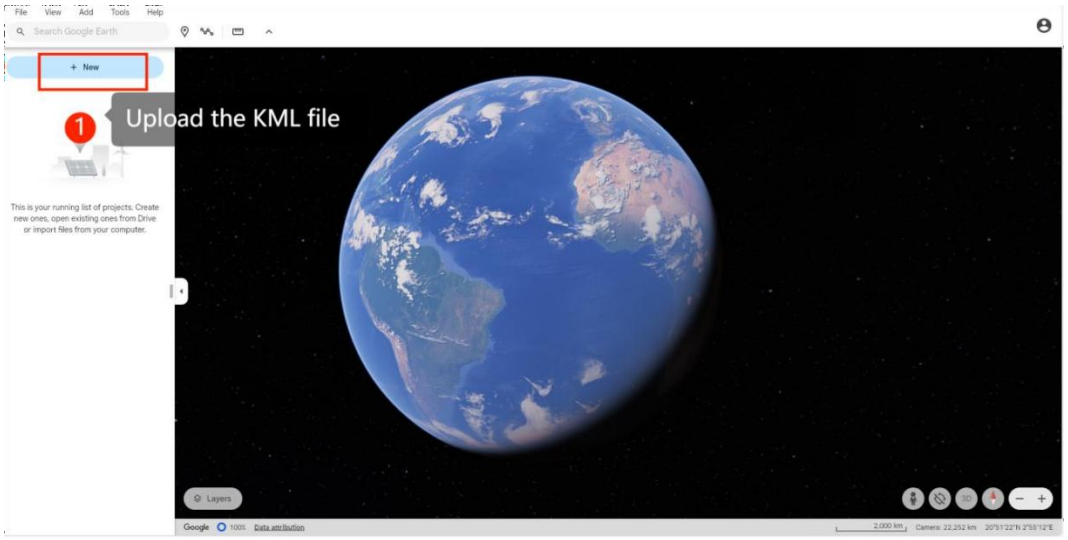


Figure 25: Illustration of google earth.

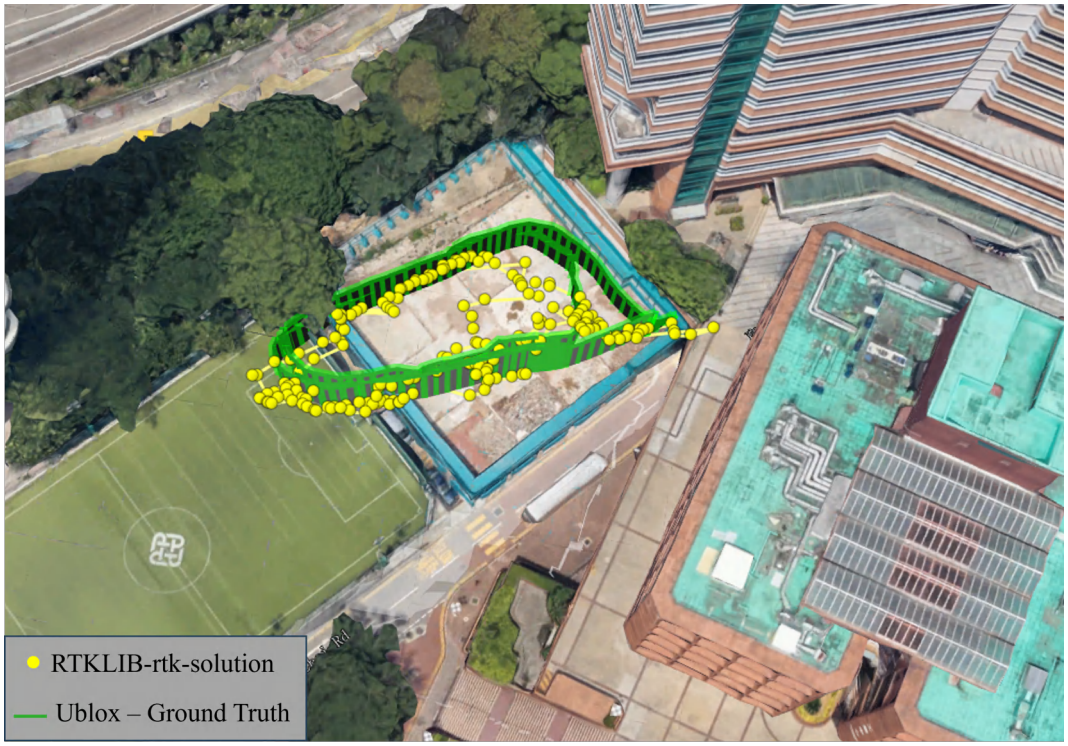


Figure 26: .kml generated on Google Map

6 RINEX Data Processing

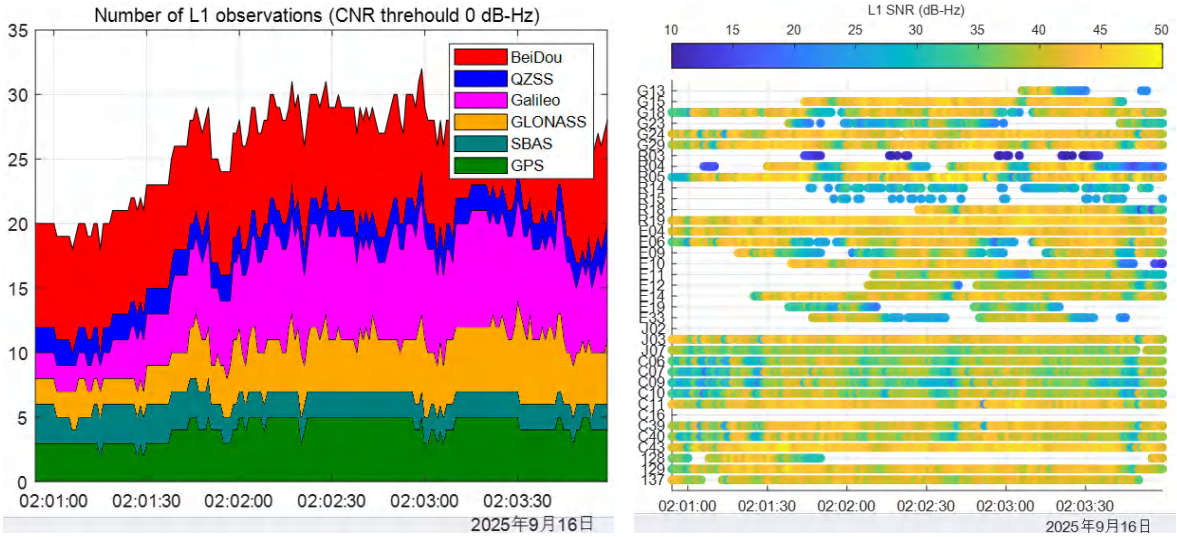
RTKLIB provides a powerful framework for comprehensive GNSS data analysis. This section demonstrates the implementation of RTKLIB functions for processing RINEX observation and navigation files, followed by detailed signal quality and satellite geometry analysis.

6.1 RINEX Data Processing

The MATLAB implementation begins with reading RINEX observation and navigation files using RTKLIB functions.

6.2 Signal-to-Noise Ratio (SNR) Analysis

The L1 SNR analysis reveals the signal quality characteristics over the observation period. The SNR plot shows variations in signal strength ranging from 10 to 50 dB-Hz, with color-coded representation indicating signal quality levels.



(a).Number of L1 Observations (b).Received L1 SNR

Figure 27: MATLAB Multi-Constellation Satellite Visibility and SNR Analysis

The analysis shows:

- High SNR values (45-50 dB-Hz) indicate excellent signal reception
- Medium SNR values (30-40 dB-Hz) represent good signal quality
- Low SNR values (10-25 dB-Hz) suggest potential multipath or obstruction effects

6.3 Multi-Constellation Satellite Visibility

The satellite visibility analysis demonstrates the contribution of different GNSS constellations throughout the observation period:

Table 4: Multi-constellation satellite visibility statistics

Constellation	Max Satellites	Avg Contribution	Color Code
BeiDou	12-15	35%	Red
GPS	8-10	25%	Green
QZSS	2-3	15%	Blue
Galileo	4-6	15%	Magenta
GLONASS	3-5	10%	Orange

6.4 Sky Plot and Satellite Geometry

The sky plot visualization provides critical insights into satellite geometry and potential dilution of precision (DOP) effects. The plot shows satellite positions in polar coordinates with azimuth (0°-360°) and elevation angles.

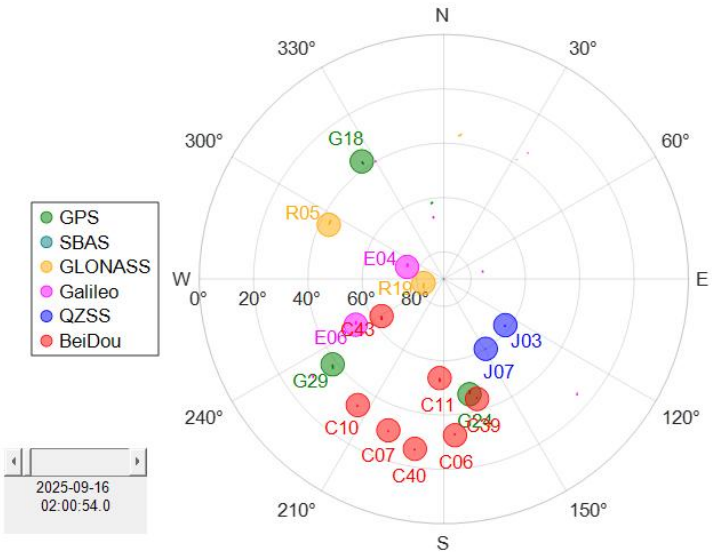


Figure 28: Sky Plot Visualization

Key observations include:

- Satellites distributed across different elevation angles from 20° to 80°
- Good azimuthal distribution ensuring geometric diversity
- BeiDou constellation provides dominant coverage in the Asian region

6.5 Data Export and Integration

The MATLAB analysis generates CSV files for further processing and integration with Python algorithms:

- pseudoranges_meas.csv - Observed pseudorange measurements of satellites
- satellite_clock_bias.csv - Computed satellite clock bias
- `satellite_positions.csv` - Computed satellite positions
- `ionospheric_delay.csv` - Modeled ionospheric corrections
- `tropospheric_delay.csv` - Modeled tropospheric corrections

7 Acquiring Ground Truth Data from u-blox

This section details the process of extracting high-precision positioning data from u-blox receivers, which is essential for serving as **ground truth** in accuracy assessments.

7.1 Rationale for u-blox Trajectory Accuracy

The trajectory data logged by high-end u-blox receivers is considered highly accurate, making it suitable for ground truth generation. This superior precision stems from the integration of several advanced technologies:

- **Multi-Mode GNSS Reception:** The receivers process signals from multiple satellite constellations (**GPS, GLONASS, Galileo, and BeiDou**), significantly increasing the number of visible satellites for better accuracy and reliability, especially in challenging environments.
- **Real-Time Kinematic (RTK) Technology:** By utilizing error correction data provided by a nearby reference station, RTK eliminates signal errors, enabling **centimeter-level position accuracy**.
- **Integrated Inertial Navigation (INS):** The fusion of GNSS data with inertial sensor information provides short-term position and attitude updates. This integration enhances the **continuity and robustness** of the positioning solution, particularly during temporary GNSS outages (e.g., in tunnels or under heavy foliage).
- **Raw Data Output:** The receivers provide high-rate, high-precision messages in the proprietary **UBX format**, such as `NAV-HPPOSECEF` (High Precision Position in ECEF).

The resulting trajectory, leveraging these technologies, is therefore considered a reliable and high-fidelity baseline for evaluating other navigation systems.

7.2 UBX to CSV Conversion for Ground Truth

To utilize the raw UBX data, it must be converted into a readable format, such as CSV. The `pyubx2` library provides an efficient tool for this conversion.

1. **Install Dependencies**
   - Ensure the `pyubx2` library is installed:

     ```
     pip install pyubx2
     ```

2. **Run GUI Converter**
   - Execute the Python GUI converter script:

     ```
     python pyubx2_csv_converter_gui.py
     ```

3. **Using the GUI**
   - **Select Input UBX File:** Browse and select your raw UBX data file.
   - **Choose Output Directory:** Select the destination folder where the converted CSV files will be saved.
   - **Start Conversion:** Click "Start" to begin the conversion process.

   The GUI offers a user-friendly experience by providing:
   - Real-time progress indication.
   - Background processing to maintain UI responsiveness.
   - Robust error handling and status updates.

4. **Key Output: Ground Truth Data**
   - The conversion process generates a critical file: `NAV-HPPOSECEF.csv`. This file contains the decoded high-precision positioning data (Earth-Centered, Earth-Fixed coordinates) that serves as the **ground truth** for subsequent accuracy assessments and comparison.

8 Python WLSE-SPP Implementation

The Python implementation focuses on advanced positioning algorithms using Weighted Least Squares Estimation for Single Point Positioning. This approach provides robust position estimation with comprehensive error analysis and statistical validation.

8.1 Algorithm Implementation Overview

The WLSE-SPP algorithm processes GNSS observations through the following key components:

1. Pseudorange observation preprocessing
2. Satellite position computation from ephemeris data
3. Atmospheric error modeling (ionospheric and tropospheric delays)
4. Weighted least squares position estimation
5. Statistical validation and accuracy assessment

8.2 Trajectory Analysis Results

The trajectory analysis demonstrates the algorithm's performance in both geographic and local coordinate systems:

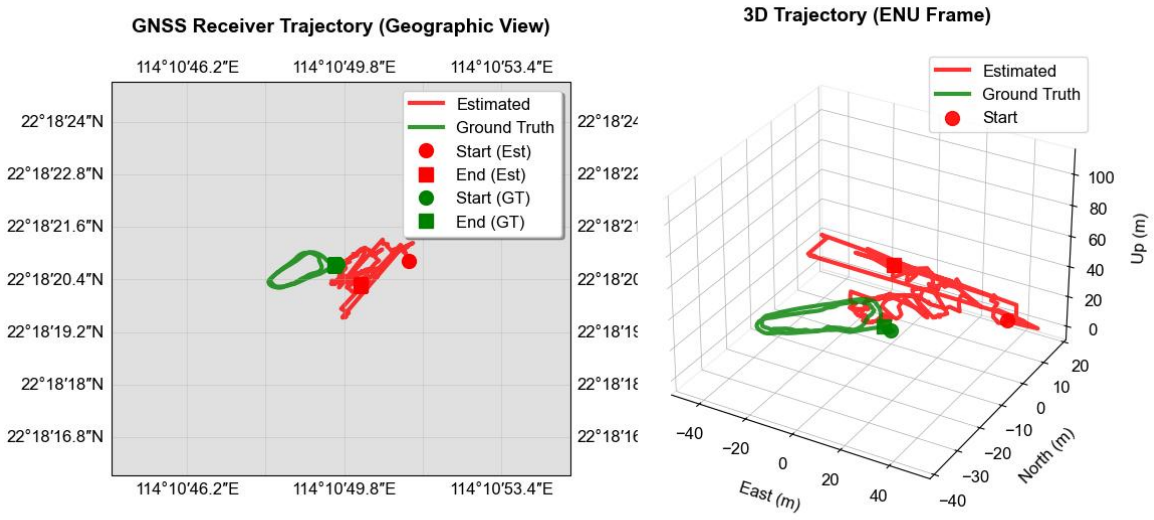


Figure 29: Python WLSE-SPP Trajectory Analysis (Geographic View and 3D ENU Frame)

8.2.1 Geographic Trajectory Visualization

The geographic view presents the estimated trajectory compared to ground truth data over the coordinate range:

- Latitude Range: 22.305614° to 22.305839° (approximately 55 meters)
- Longitude Range: 114.180019° to 114.180466° (approximately 45 meters)
- Altitude Range: 53.697 m to 135.184 m

8.2.2 3D ENU (East-North-Up) Frame Analysis

The 3D trajectory analysis in the ENU coordinate system reveals the positioning behavior in local coordinates, with distinct start and end points clearly marked. The trajectory shows convergence behavior typical of static positioning scenarios.

8.3 Comprehensive Error Analysis

The error analysis provides detailed insights into positioning performance across different dimensions and temporal characteristics.

8.3.1 Directional Positioning Errors

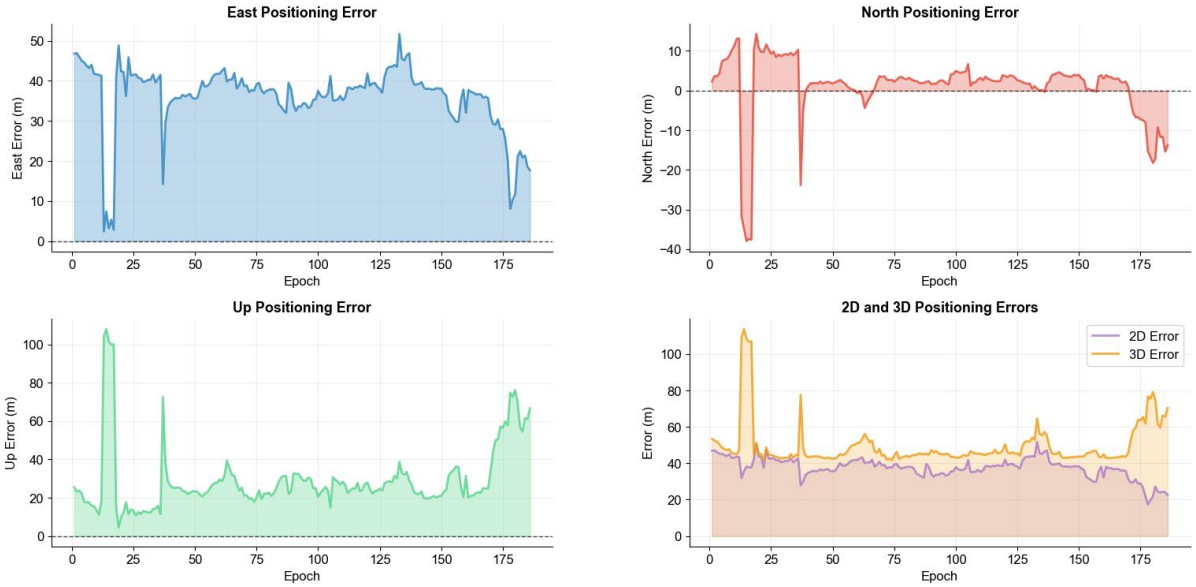


Figure 30: Directional Positioning Errors Analysis (East, North, Up, and 2D/3D Errors)

Individual error components demonstrate different characteristics:

Table 5: Directional error characteristics

Error Component	Characteristics	Typical Range
East Error	Decreasing trend over epochs	5-50 meters
North Error	Relatively stable with occasional excursions	-30-10 meters
Up Error	High variability, characteristic altitude challenges	0-100 meters

8.3.2 Statistical Error Distribution Analysis

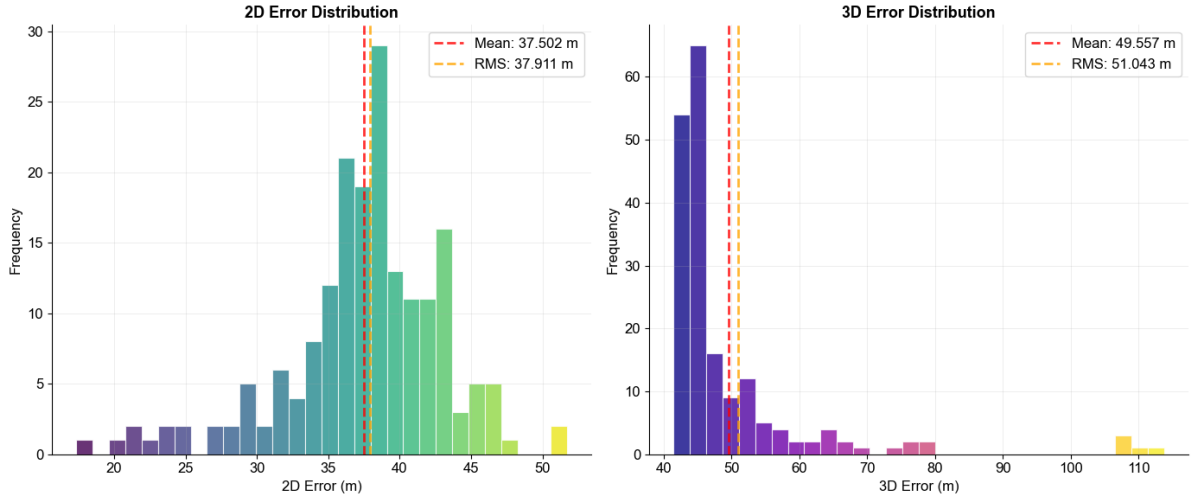


Figure 31: Statistical Error Distribution Analysis (2D and 3D Error Histograms)

The comprehensive statistical analysis reveals important positioning characteristics:

Table 6: Statistical error summary

Error Type	Mean (m)	RMS (m)
2D Error	37.502	37.911
3D Error	49.557	51.043

8.4 Error Evolution and Convergence Analysis

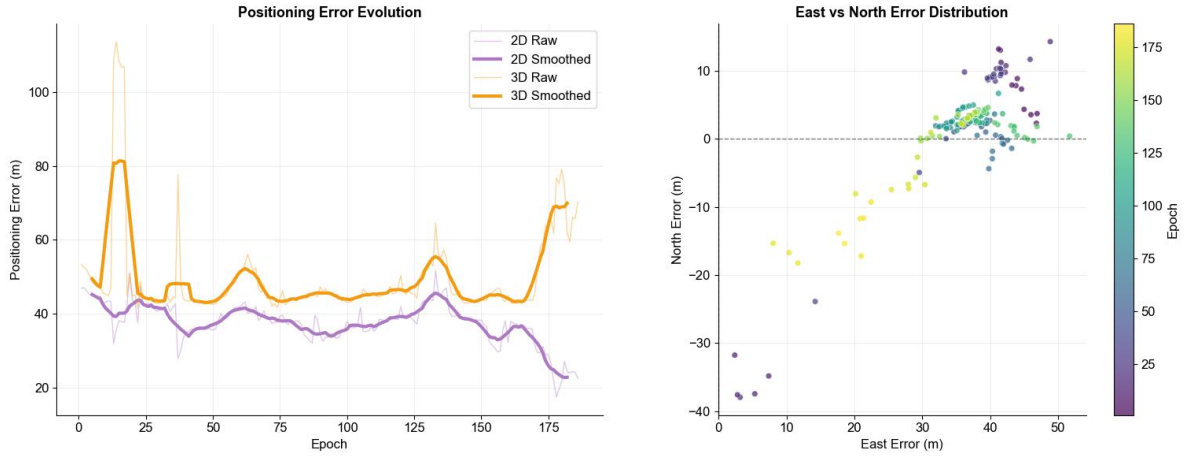


Figure 32: Error Evolution Analysis and East vs North Error Distribution

The positioning error evolution analysis demonstrates the convergence characteristics of the WLSE-SPP algorithm:

- Initial convergence period showing higher errors in the first 25 epochs
- Stabilization phase from epochs 25-150 with consistent performance
- Final degradation period potentially due to satellite geometry changes

8.5 Comprehensive Positioning Accuracy Statistics

The final analysis summary provides complete positioning accuracy statistics for the 186 processed epochs:

```
============================================================
        GNSS SPP POSITIONING RESULTS SUMMARY
============================================================
Total Epochs Processed: 186
============================================================
          POSITIONING ACCURACY STATISTICS
============================================================
Metric              2D Error (m)    3D Error (m)
------------------------------------------------------
Mean                37.502          49.557
RMS                 37.911          51.043
Standard Deviation  5.553           12.227
Maximum             51.699          113.729
Minimum             17.350          41.472
95th Percentile     45.208          73.716
============================================================
                COORDINATE RANGES
============================================================
Latitude Range:  22.305424°  to 22.305920°
Longitude Range: 114.180416°  to 114.180938°
Altitude Range:  10.550 m to 114.610 m
============================================================
```

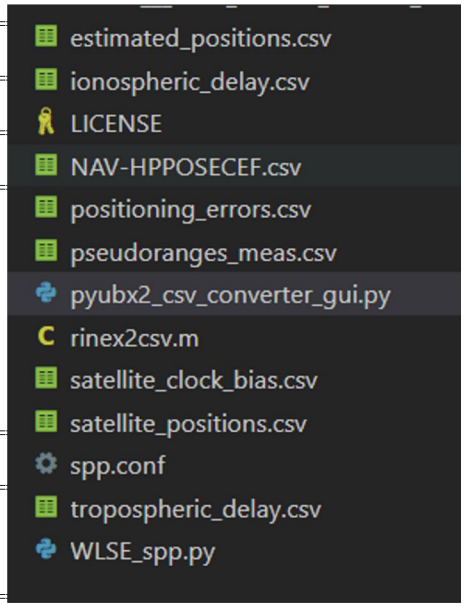


Figure 33: GNSS SPP Positioning Results Summary and Generated Output Files

Table 7: Comprehensive positioning accuracy statistics

Metric	2D Error (m)	3D Error (m)
Mean	37.502	49.557
RMS	37.911	51.043
Standard Deviation	5.553	12.227
Maximum	51.699	113.729
Minimum	17.350	41.472
95th Percentile	45.208	73.716

8.6 Output Data Generation

The Python WLSE-SPP implementation generates comprehensive output files for further analysis:

- `estimated_positions.csv` - Final position estimates for each epoch
- `positioning_errors.csv` - Detailed error analysis by epoch

9 Conclusions

9.1 Key Learning Outcomes

Through this comprehensive laboratory exercise, students achieve the following learning outcomes:

1. Understanding of multi-constellation GNSS signal characteristics and quality assessment
2. Practical experience with professional GNSS processing software (u-center) and custom algorithm implementation
3. Comprehensive error analysis techniques and statistical validation methods
4. Comparative evaluation of different positioning approaches and their trade-offs
5. Data integration and visualization skills across multiple software platforms

9.2 Recommendations for Future Work

Based on the analysis results, several areas for improvement and extension are identified:

- Implementation of advanced multipath mitigation techniques
- Integration of precise orbit and clock corrections
- Development of real-time positioning capabilities
- Extended urban canyon environment testing
- Integration with u-blox GNSS services such as AssistNow for improved performance

References

[1] T. Takasu, *RTKLIB ver.2.4.2 Manual*, 2013. Official RTKLIB documentation (SPP/RTK algorithms and configuration guidelines). https://www.rtklib.com/rtklib_document.htm.

[2] T. Takasu, *RTKLIB — Open Source Program Package for GNSS Positioning*, GitHub repository, https://github.com/tomojitakasu/RTKLIB.

[3] E. D. Kaplan and C. J. Hegarty (eds.), *Understanding GPS/GNSS: Principles and Applications*, 3rd ed., Artech House, 2017.

[4] European Space Agency (ESA), *GNSS Data Processing, Volume I: Fundamentals and Algorithms*, ESA TM-23/1, 2013. Available via ESA Navipedia.